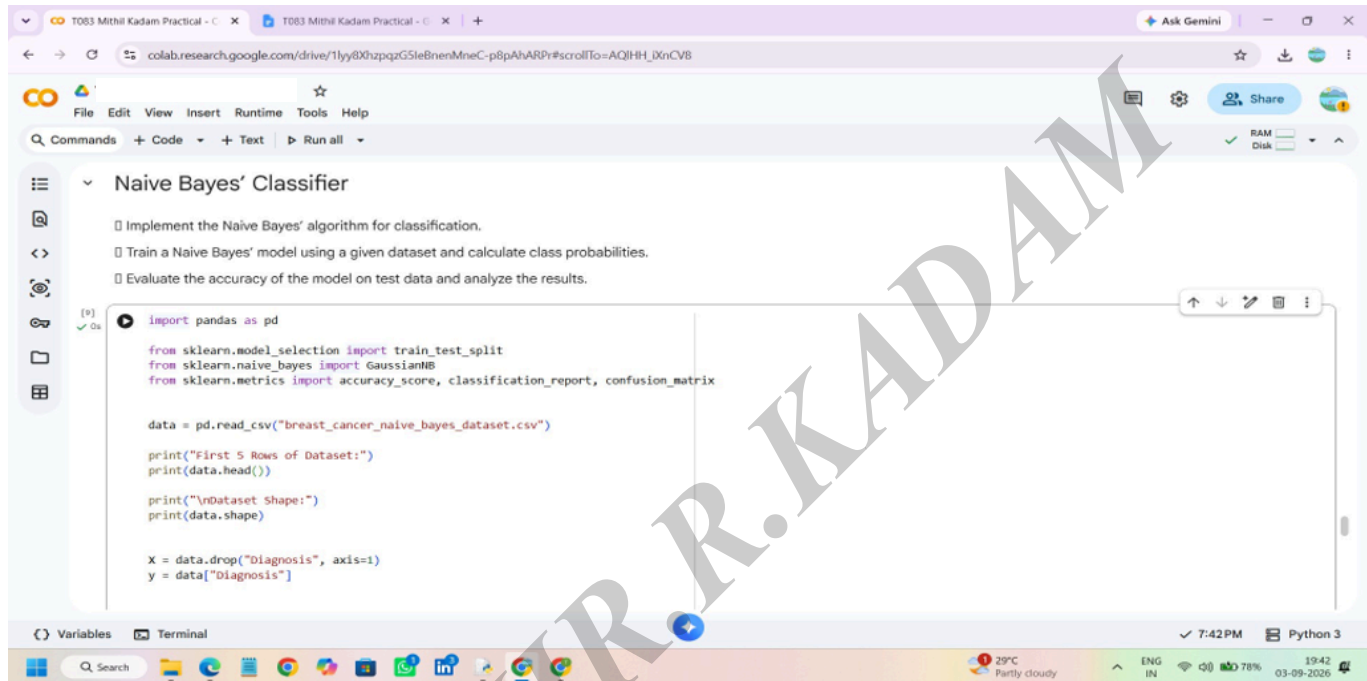


Aim:- Naive Bayes' Classifier

Implement the Naive Bayes' algorithm for classification.

Train a Naive Bayes' model using a given dataset and calculate class probabilities.

Evaluate the accuracy of the model on test data and analyze the results.



The screenshot shows a Google Colab notebook titled "Naive Bayes' Classifier". The code in the first cell is as follows:

```
import pandas as pd

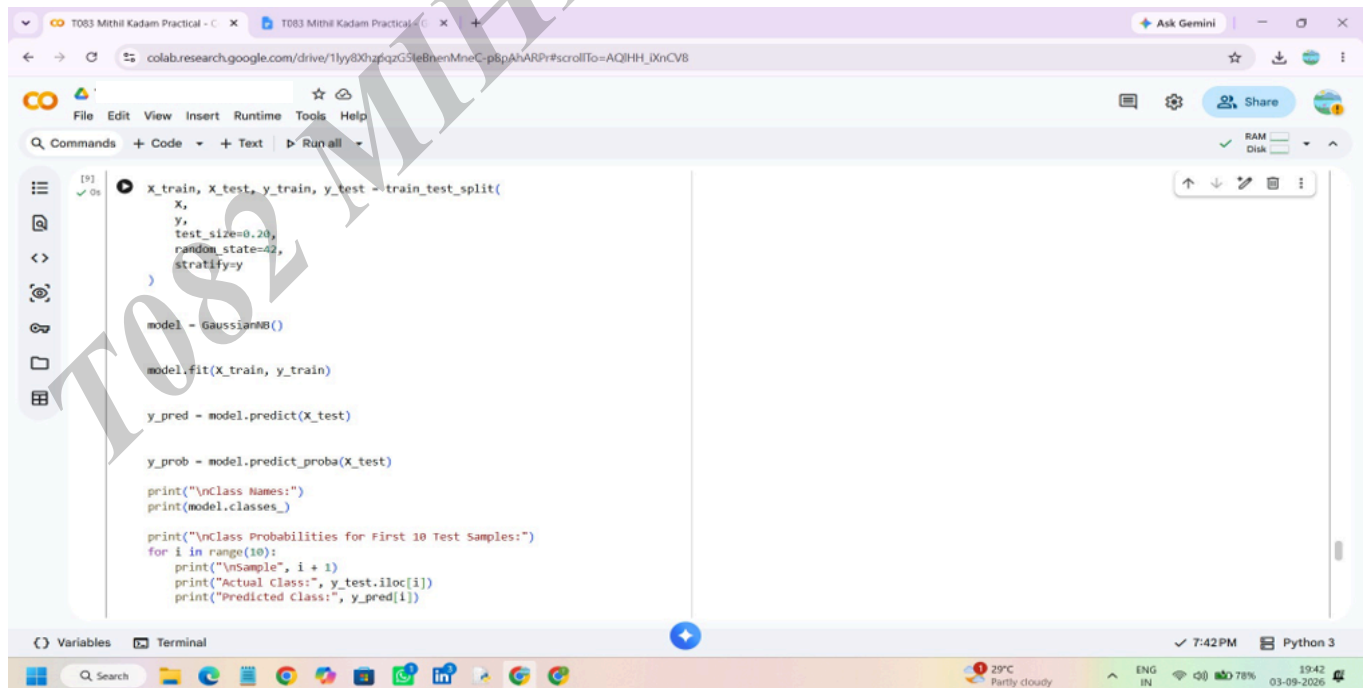
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

data = pd.read_csv("breast_cancer_naive_bayes_dataset.csv")

print("First 5 Rows of Dataset:")
print(data.head())

print("\nDataset Shape:")
print(data.shape)

X = data.drop("Diagnosis", axis=1)
y = data["Diagnosis"]
```



The screenshot shows the continuation of the code in the second cell of the Google Colab notebook. The code is as follows:

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

model = GaussianNB()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

y_prob = model.predict_proba(X_test)

print("\nClass Names:")
print(model.classes_)

print("\nClass Probabilities for First 10 Test Samples:")
for i in range(10):
    print("\nsample", i + 1)
    print("Actual Class:", y_test.iloc[i])
    print("Predicted Class:", y_pred[i])
```

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The screenshot shows a Google Colab notebook with the following code and output:

```
print("Predicted Class:", y_pred[i])

for j, class_name in enumerate(model.classes_):
    print(class_name, "Probability:", round(y_prob[i][j], 4))

accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:", round(accuracy * 100, 2), "%")

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

Output:

```
Sample 4
Actual Class: Malignant
Predicted Class: Benign
Benign Probability: 0.9968
Malignant Probability: 0.0032

Sample 5
Actual Class: Benign
Predicted Class: Benign
Benign Probability: 1.0
Malignant Probability: 0.0

Sample 6
```

The screenshot shows a Google Colab notebook displaying the first 5 rows of a dataset. The data is presented in two tables.

	mean radius	mean texture	mean perimeter	mean area	mean smoothness
0	17.99	10.38	122.80	1001.0	0.11840
1	20.57	17.77	132.90	1326.0	0.08474
2	19.69	21.25	130.00	1203.0	0.10960
3	11.42	20.38	77.58	386.1	0.14250
4	20.29	14.34	135.10	1297.0	0.10030

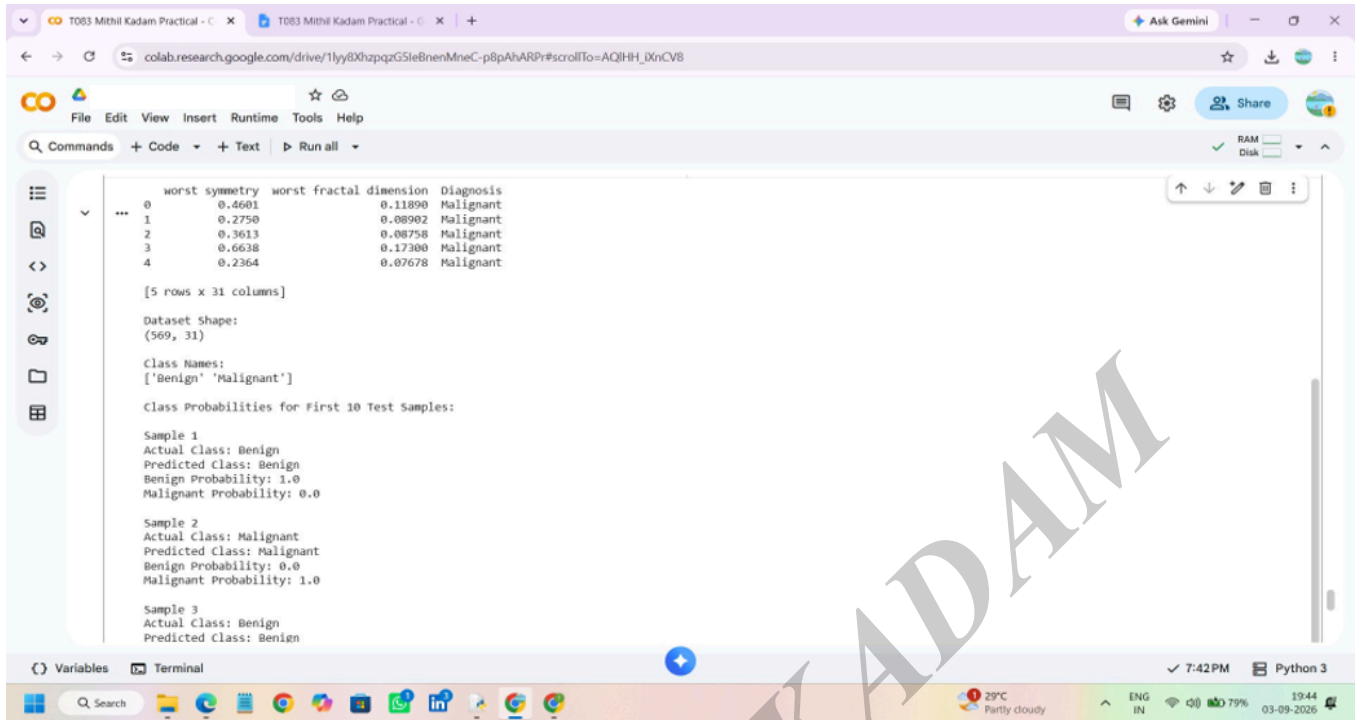
	mean compactness	mean concavity	mean concave points	mean symmetry
0	0.27760	0.1001	0.14710	0.2419
1	0.07864	0.0869	0.07017	0.1812
2	0.15990	0.1974	0.12790	0.2069
3	0.28390	0.2414	0.10520	0.2597
4	0.13280	0.1580	0.10430	0.1809

	mean fractal dimension	worst texture	worst perimeter	worst area
0	0.07871	17.33	184.60	2019.0
1	0.05667	23.41	158.80	1956.0
2	0.05999	25.53	152.50	1709.0
3	0.09744	26.50	98.87	567.7
4	0.05883	16.67	152.20	1575.0

	worst smoothness	worst compactness	worst concavity	worst concave points
0	0.1622	0.6656	0.7119	0.2654
1	0.1238	0.1866	0.2416	0.1860
2	0.1444	0.4245	0.4504	0.2430
3	0.2098	0.8663	0.6869	0.2575
4	0.1374	0.2050	0.4000	0.1625

	worst symmetry	worst fractal dimension	Diagnosis
0	0.4601	0.11890	Malignant
1	0.2750	0.08902	Malignant

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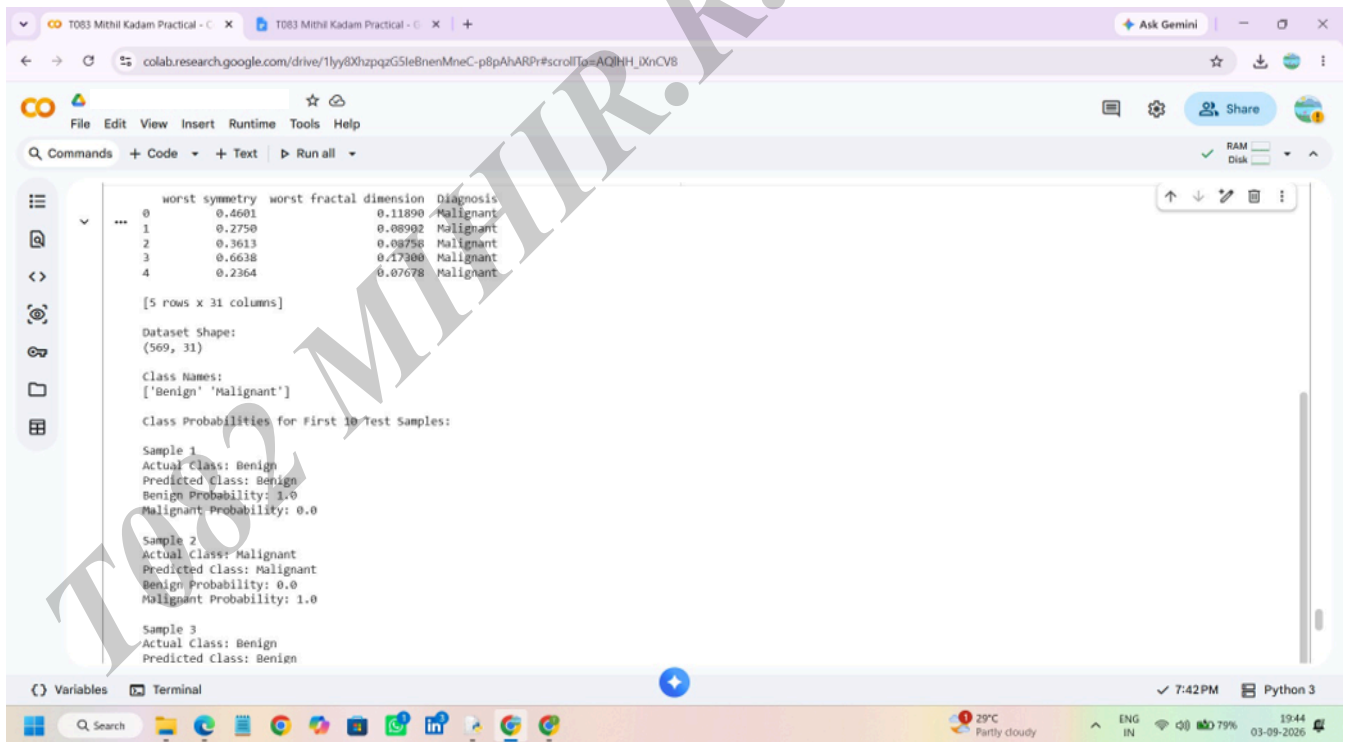
The screenshot shows a Google Colab environment with a Jupyter Notebook. The notebook contains a table of breast cancer data and a summary of the dataset. The table has columns for 'worst symmetry', 'worst fractal dimension', and 'Diagnosis'. The data is as follows:

	worst symmetry	worst fractal dimension	Diagnosis
0	0.4601	0.11890	Malignant
1	0.2750	0.08902	Malignant
2	0.3613	0.08758	Malignant
3	0.6638	0.17300	Malignant
4	0.2364	0.07678	Malignant

Below the table, the notebook displays the following information:

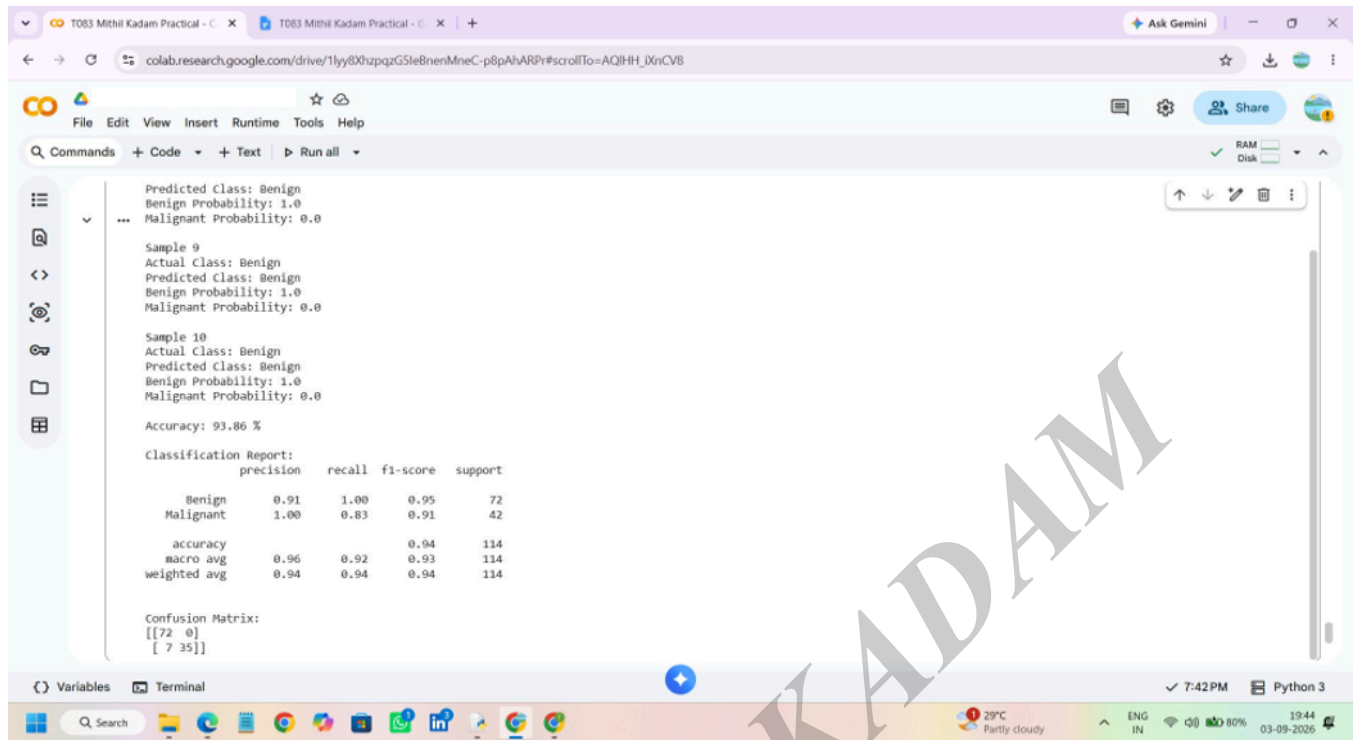
- [5 rows x 31 columns]
- Dataset Shape: (569, 31)
- Class Names: ['Benign', 'Malignant']
- Class Probabilities for First 10 Test Samples:
- Sample 1: Actual Class: Benign, Predicted Class: Benign, Benign Probability: 1.0, Malignant Probability: 0.0
- Sample 2: Actual Class: Malignant, Predicted Class: Malignant, Benign Probability: 0.0, Malignant Probability: 1.0
- Sample 3: Actual Class: Benign, Predicted Class: Benign

The interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for file operations, and a status bar at the bottom showing the time (7:42 PM) and Python 3 environment.



This is a duplicate of the screenshot above, showing the same Google Colab environment with the Jupyter Notebook displaying the breast cancer dataset analysis.

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```
Predicted Class: Benign
Benign Probability: 1.0
Malignant Probability: 0.0

Sample 9
Actual Class: Benign
Predicted Class: Benign
Benign Probability: 1.0
Malignant Probability: 0.0

Sample 10
Actual Class: Benign
Predicted Class: Benign
Benign Probability: 1.0
Malignant Probability: 0.0

Accuracy: 93.86 %

Classification Report:
              precision    recall  f1-score   support

   Benign         0.91         1.00         0.95         72
  Malignant         1.00         0.83         0.91         42

   accuracy          0.96          0.92          0.93        114
  macro avg          0.96          0.92          0.93        114
 weighted avg          0.94          0.94          0.94        114

Confusion Matrix:
[[72  0]
 [ 7 35]]
```

The screenshot shows a Google Colab notebook interface. The top bar includes the Google Colab logo, file name 'T083 Mithi Kadam Practical', and a 'Share' button. The left sidebar contains icons for file management, search, and other tools. The main area displays the output of a classification model. The output includes predicted and actual classes, probabilities, and a confusion matrix. A large watermark 'T082 MIHIR.R.KADAM' is visible across the image.

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